

Exercise 4:

Modify and optimize the Pintool **bprofile.cpp** shown in class and located in the moodle site at <https://moodle25.technion.ac.il/mod/resource/view.php?id=256169>

to:

1. Successfully run on all the provided input binaries from exercise 3.
2. Avoid saving and restoring “dead” registers that are used in the profiling instrumentation code.
3. Generate the output profiling file “**edge-profile.csv**” instead of the currently generated file “**bprofile.out**” in the following format of exercise 3.

<bbl₁ address>, <bbl₁ exec count>, <taken count>, <fallthru count>, <exec count of each indirect target
jump up to 4 target addresses: <target addr₁, exec count>, <target addr₂, exec count>, ..., <target addr₄,
exec count>>

<bbl₂ address>, <bbl₂ exec count>, <taken count>, <fallthru count>, <exec count of each indirect target
jump up to 4 target addresses : <target addr₁, exec count>, <target addr₂, exec count>, ..., <target addr₄,
exec count>>

...

<bbl_n address>, <bbl_n exec count>, <taken count>, <fallthru count>, <exec count of each indirect target
jump up to 4 target addresses:: <target addr₁, exec count>, <target addr₂, exec count>, ..., <target addr₄,
exec count>>

The basic blocks should be **sorted** from most frequently executed (“hottest”) to the least frequently executed (“coldest”) ones.

You can assume that the total number of basic blocks is less than 10,000.

4. Identify cases of large differences between the generated profiling and the generated profiling from exercise 3 and add the reason in the exercise README file.
5. The optimized pintool should show at least 5% improvement in one of the following performance measures over the provided non-optimized version, when executed with a specified profiling interval using the flag **-prof_time <profiling interval in seconds>**:
 - a. elapsed running time
 - b. user time
 - c. system time
 - d. number of page faults
 - e. total number of cycles

Submission requirements:

The submission of this exercise is **in pairs only**.

Submit 1 compressed file called “**ex4.zip**” into the moodle exercise 3 [link](#) containing the following files:

1. The modified binary **bprofile.so** (compiled and tested by you that it runs and gives the result).
2. A directory called: ‘src’ containing all the sources of your pintool along with the make files ‘**makefile**’, ‘**makefile.rules**’ and a **README.txt** file that includes the following:
 - a. names + id numbers
 - b. How to run the tool.
 - c. What problems did you fix in the pintool and how did you choose to fix them.
 - d. What were the large differences between the generated profiling information in this exercise and the one collected in exercise 3 and why do you think the happen.

Submission deadline is Sunday, July 19, 2026 at midnight.